

The Canva logo is displayed in white script font on a blue-to-purple gradient rounded square. The background features a light blue gradient with dark blue and black diagonal stripes.

Canva

Cyber Forensic Investigation

Canva Data Breach 2019

Executive Summary

Overview:

- In May 2019, Canva suffered a major data breach affecting ~139 million users.

Key Findings:

- Breach due to credential stuffing.
- Compromised data: names, emails, hashed passwords.

Tools Used:

- Volatility (Memory analysis)
- Wireshark (Network monitoring)
- Splunk (Log correlation)
- Autopsy (File system review)

Recommendations:

- Enforce MFA
- Zero Trust model
- Stronger hashing algorithms

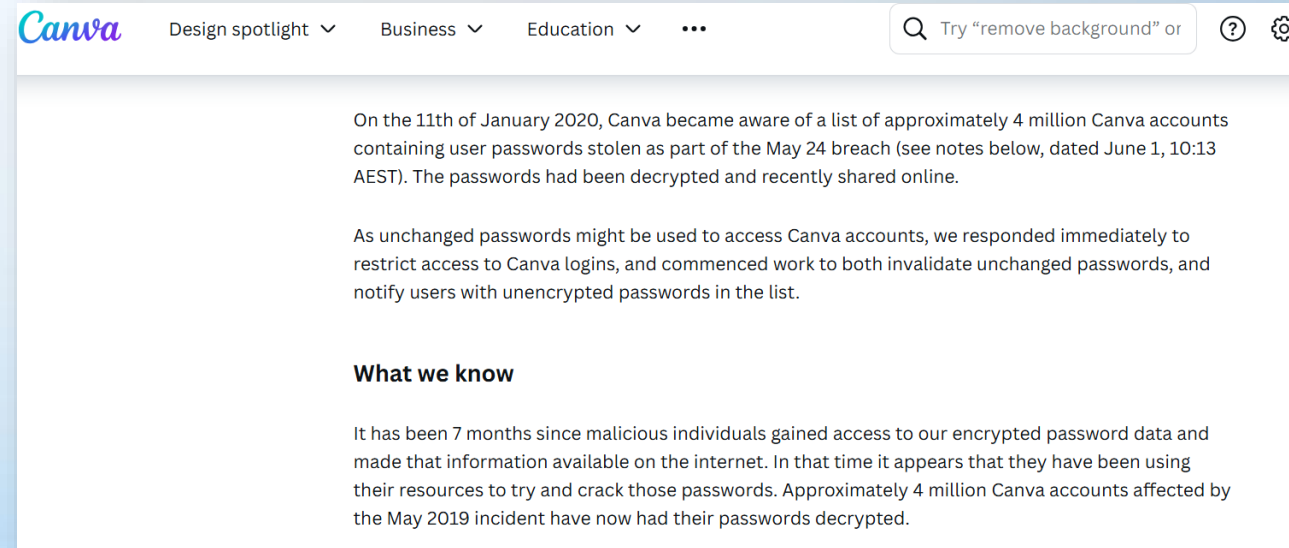
Introduction

- **Significance:** One of the largest breaches in Australian tech history.
- **Scope:** Investigating breach source, impact, and forensics response.
- **Methodology:**
 - Memory and file system analysis
 - Log and network traffic investigation
- **Tools:** Volatility, Wireshark, Splunk, Autopsy



Incident Overview

- **Timeline:**
 - Breach discovered: May 24, 2019
 - Public disclosure: Same day
- **Impact:**
 - 139 million users impacted
- **Attack Vector:**
 - Credential stuffing via reused passwords
- **Data Compromised:**
 - Full names, emails, bcrypt-hashed passwords



Forensic Evidence Collection

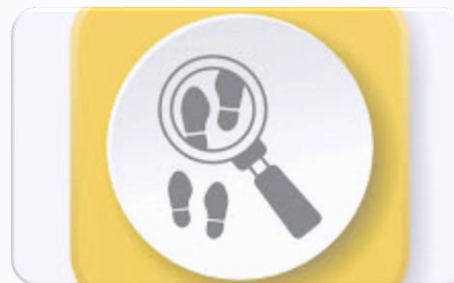
- **Steps Taken:**
 - System isolation
 - Evidence preservation (chain of custody)
- **Data Acquisition:**
 - Disk images, volatile memory
 - Checksum verification (MD5/SHA-1)
- **Analysis Tools:**
 - **Volatility:** Memory dump analysis
 - **Autopsy:** File system and metadata examination



Network Traffic & SIEM Analysis



splunk>



Wireshark:

- Identified suspicious outbound connections (e.g., to known blacklisted IPs)

Splunk:

- Correlated user login anomalies, failed logins, access times

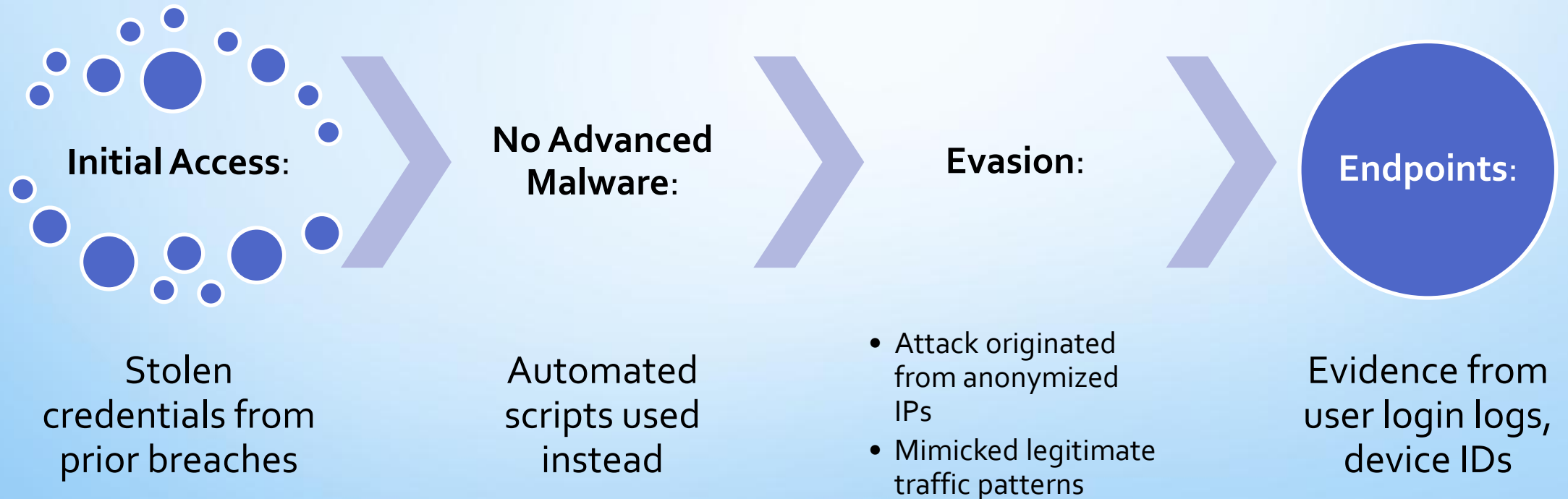
IoCs:

- IPs, login patterns, access timestamps

Detection:

- Evidence of repeated credential stuffing attempts

Attack Methodology



Legal & Compliance Analysis



Australian Privacy Act 1988:

- Breach of data security principles



GDPR Implications:

- EU user data affected — raised cross-border compliance issues



Compliance Failures:

- Delay in full technical disclosure
- Lacked proactive breach mitigation controls

Strategic Recommendations

- **Zero Trust Architecture:**
 - No implicit trust; verify everything
- **MFA Enforcement:**
 - Especially for admin and high-privilege accounts
- **Password Hygiene:**
 - Routine audits; enforce unique, complex passwords
 - Upgrade to bcrypt with higher cost factor
- **SIEM Monitoring:**
 - Real-time detection of anomalies using Splunk/ELK



Practical Demonstration

- Volatility
- Autopsy

The screenshot displays a practical demonstration of digital forensics tools. It is divided into three main sections:

- Top Left (Terminal):** Shows the execution of the Volatility Framework. The command `python2 vol.py -f 0zapftis.vmem imageinfo` is run. The output indicates that the profile is determined based on a KDBG search, suggesting a WinXPSP2x86 profile. It also lists system information such as the number of processors (1), image type (Service Pack), and image date and time (2011-10-10 17:06:54 UTC+0000).
- Top Right (Terminal):** Shows the execution of the Volatility Framework's `pslist` command. The output is a table listing running processes, including System, smss.exe, csrss.exe, winlogon.exe, services.exe, lsass.exe, vmacthlp.exe, svchost.exe, spoolsv.exe, VMWareService.e, alg.exe, wscntfy.exe, explorer.exe, VMwareTray.exe, and VMWareUser.exe. Each entry includes details like PID, PPID, Thds, Hnds, Sess, Wow64, Start, and Exit.
- Bottom Left (Terminal):** Shows the execution of the Autopsy Forensic Browser. The command `autopsy` is run. The output displays the Autopsy Forensic Browser interface, including the URL `http://www.sleuthkit.org/autopsy/` and the version (2.24). It also shows the evidence locker path (`/var/lib/autopsy`), start time (Sun May 11 00:20:36 2025), remote host (localhost), and local port (9999).
- Bottom Right (Web Browser):** Shows the Autopsy Forensic Browser interface. The browser displays a warning that Java Script is enabled. Below the warning, it shows the Autopsy Forensic Browser 2.24 logo and a cartoon dog. The browser also displays search results for ASCII and Unicode, indicating 205295 hits for ASCII and 437 hits for Unicode. The search options are set to Case Sensitive.